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PERSONAL SAVINGS IN THE POSTWAR WORLD: IMPLICATIONS FOR THE THEORY OF HOUSEHOLD BEHAVIOR

Towards a Theory of Saving Behavior

By F. THOMAS JUSTER AND LESTER D. TAYLOR*

The behavior of personal saving in the *U.S.* economy has long held a fascination for economists. The earliest consumption functions, based on cross-sectional data from household budget surveys, indicated that richer households save more relative to income than poorer ones, leading to the Keynesian speculation that secularly rising income would result in higher saving rates. The monumental study of *U.S.* capital formation of Simon Kuznets indicated that this attractive theory did not square with the facts, since there was no evidence that saving as a fraction of income increased as income rose secularly.

This apparent contradiction between cross-sectional and long-term time-series evidence attracted a great deal of attention in the late 1940's and early 1950's and led to the formulation of three new theories of the consumption function, the relative income hypothesis of J. S. Duesenberry, the life-cycle hypothesis of F. Modigliani and R. Brumberg, and the permanent-income hypothesis of Milton Friedman.¹ Although these theories, especially the last two, have inspired a huge literature, its main strands really have had little to say about cyclical movements in personal saving, as opposed to secular movements.²

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¹ A fourth theory, developed by James Tobin, involves an extension of the absolute-income hypothesis of Keynes that takes into account the household's level of wealth.

² For a detailed review of the permanent-income

While focusing on the secular behavior of saving was understandable in the 1950's and early 1960's when the personal saving rate moved within fairly narrow limits, the saving rate in recent years has been considerably more volatile. This cyclical volatility has been a concern of both Juster and Taylor in recent years, and the present undertaking represents a merging of efforts. The focus of Taylor's work has been on the role of income composition in determining the behavior of aggregate saving, while Juster has concentrated on the role of uncertainty.³ The present paper attempts to combine both of these insights into a more refined model of personal saving.

I. Income Composition and Adjustment Effects

Excluding the effects of uncertainty, which will be discussed below, the general framework for our analysis is the "zero-depreciation" theory of saving elaborated by H. S. Houthakker and Taylor. According to this theory, personal saving (S) is assumed to be a linear function of the existing stock of financial assets (A) and income (Y):

$$(1) \quad S(t) = \alpha + \beta A(t) + \gamma Y(t).$$

The stock of financial assets is assumed to

hypothesis and critical evaluation of the many and various attempts to test it, see T. Mayer.

³ See Taylor (1971 and 1974). Also, see Juster (1973) and Juster and P. Wachtel (1972a and 1972b).

be nondepreciating, with the result that the relationship between the stock A and its flow S is given by

$$(2) \quad \dot{A}(t) = S(t).$$

The basic notion underlying the model is that the consumer adjusts saving to bring the stock of financial assets into line with income. This model can be interpreted as embodying a Stone-Nerlove stock adjustment process to the stock of wealth, with the proviso that the depreciation rate on the stock is zero. An important consequence of this assumption is that saving is zero in long-run, steady-state equilibrium. Accordingly, the concept of a marginal propensity to save must be with reference to some limited interval of time. More particularly, γ in equation (1) refers to the instantaneous marginal propensity to save defined by the condition that the underlying stock of financial assets is constant. However, γ should not be confused with the one-quarter marginal propensity to save which will be the focus in the empirical work.

Following Taylor (1971), one of the objectives of our inquiry requires disaggregation of disposable income into the following components: labor income (L), property income (PR), transfers (TR), personal contributions to social insurance (SI), and personal tax and nontax payments (T). With this decomposition of disposable income, equation (1) becomes

$$(3) \quad S(t) = \alpha + \beta A(t) + \gamma_1 L(t) + \gamma_2 PR(t) + \gamma_3 TR(t) + \gamma_4 SI(t) + \gamma_5 T(t).$$

Since savings are the epitome of a durable good, β should be negative: the stock of wealth should have the same (qualitative) effect on saving that the stock of automobiles, say, has on the sale of new cars. The coefficients γ_1 , γ_2 , and γ_3 should, of course, be positive and on theoretical grounds γ_2 should be greater than γ_1 be-

cause property income is more uncertain than labor income. The marginal propensity to save out of transfer payments has traditionally been thought to be low, but all empirical work to date with this model has found γ_3 to be in the neighborhood of .8. Finally, γ_4 and γ_5 are naturally expected to be negative, and since contributions to social insurance leave the contributors with a claim to future income—rather than just a tax receipt—the size of γ_4 will depend upon the extent to which social insurance substitutes for regular saving. At an extreme, γ_4 could be in the neighborhood of -2 , since employees' contributions are matched dollar-for-dollar by employers.

As formulated in (3), the model contains a quantity, the stock of financial assets, that can at best be measured with difficulty, hence it is desirable that it be eliminated. Once this is done, and also after the model is translated into one involving discrete intervals of time, the estimating form of the model, with an error term (u) added, becomes:⁴

$$(4) \quad S_t = B_1 S_{t-1} + B_2 \Delta L_t + B_3 \Delta PR_t + B_4 \Delta TR_t + B_5 \Delta SI_t + B_6 \Delta T_t + u_t.$$

The coefficients $B_1, \dots, B_6$ are functions of the structural parameters $\beta, \gamma_1, \dots, \gamma_5$ (α is indeterminate), and estimates of the latter can be readily obtained from estimates of the former.

II. The Role of Uncertainty

Economists have long paid lip service to the role of uncertainty in affecting economic behavior, but attempts to quantify the impact of uncertainty have been few and far between. Empirically, uncertainty

⁴ The details of this elimination and translation are set out in Houthakker and Taylor, pp. 13–17. For the formulae involved, see p. 17.

regarding income usually takes the form of supposing that people make judgments about the future on the basis of past behavior of the quantities in question. However, for reasons that need not be discussed at length, the customary procedures seem to us to be seriously deficient, and we feel that there may be some real payoff in devising direct measures of uncertainty that are both relevant and operational.

Following Juster and P. Wachtel (1972a) and Juster (1973), our approach takes into account two types of uncertainty that are expected to influence saving behavior. The first type is the conventional kind of uncertainty regarding income. If consumers become concerned that money income may fall, they can be expected to react by trying to build up cash balances via increased saving. If money income actually falls, the fixed-cost elements of consumption expenditures can be expected to result in lower saving. Empirical representation of these effects can be accomplished by using the unemployment rate to measure money income that is high or low relative to the recent past, and to use the change in unemployment rate as a measure of changes in the degree to which consumers attach uncertainty to the receipt of future money income. Thus, unemployment is expected to have a negative impact on current saving, since it represents a negative transitory income effect, while the change in unemployment is expected to have a positive impact on saving.

The second, less conventional, and perhaps more important, dimension of uncertainty has to do with the effects of price inflation on saving. A recent paper by Wachtel (1974) suggests a useful classification for the effects of price inflation on saving: a money-illusion effect, an intertemporal substitution effect, and an uncertainty effect.

The first two of these effects are traditional subjects of analysis, but the third is a relatively recent and more speculative insight and is based on an idea long espoused by George Katona that rising pessimism regarding the future results in higher rates of saving. As elaborated by Juster (1973), the key ingredient is the downward inflexibility of money wage rates. In a world where money wage rates cannot fall, a zero rate of price inflation implies that changes in real wage rates cannot be negative. Consequently, uncertainty about the change in real income is limited to uncertainty about how large an increase in money wages will be forthcoming.

However, when prices are rising, real income can fall if money wages fail to rise as rapidly. Moreover, the higher the rate of price inflation, the greater the possible downward movement in real income. Viewed more generally, the distribution of the expected change in real wages is determined jointly by the distribution of the expected change in money wages, the distribution of the expected change in prices, and the relationship between these two quantities. Only in the extreme case of automatic and complete indexing will the existence of price inflation (of uncertain magnitude) not lead to greater uncertainty about real wages than in a world where price inflation is absent.

While price inflation *per se* is sufficient to generate uncertainty about the change in real income, the measure that best represents the kind of uncertainty that impacts on saving behavior might well be the maximum price rise considered "plausible" by consumers. That is, saving may be more influenced by judgments about plausible upper limits of future price increases than by estimates of the most probable change in prices. Although we do not have a direct measure of these upper

limits, it does not seem unreasonable to suppose that the variation among households in the expected change in prices provides some useful indirect information.⁵

III. Empirical Results

The empirical analysis employs quarterly data, mostly taken from the National Income Accounts (*NIA*), and covers the period beginning with the second quarter of 1953 and ending with the fourth quarter of 1973. The *NIA* data are aggregates, seasonably adjusted at annual rates, expressed in billions of current dollars. The price expectations data come from surveys conducted periodically by the Survey Research Center at the University of Michigan.⁶ The conventional

⁵ For example, if all households expect prices to rise at the rate of zero percent per year, this may not be very different in terms of the impact on saving than if, *ceteris paribus*, all households expect prices to rise at a rate of 10 percent per year. But if there is a great deal of dispersion in "best guesses" about the extent of future price rises, this dispersion may be taken as a reasonable proxy for concerns about plausible upper limits. In any event, this variation across households represents the only empirical measure of uncertainty that we have, except, of course, for the mean expected price change across all households.

⁶ The questions are relatively straightforward: "Do you expect prices to rise or fall or what?" and, starting with the first quarter of 1966, "How much do you expect prices to rise—1 or 2 percent, 3 or 4 percent, 5 percent, between 5 and 10 percent or more?" Estimates of the mean and variance of expected price change among the population sampled can be constructed either by assignment of quantitative magnitudes to basically qualitative responses for part of the period (i.e., until 1966) and assignment of midpoints for that part of the period for which quantitative data are available or by a less arbitrary, but perhaps also less satisfactory, procedure of throwing away all quantitative detail and focusing simply on responses in the up, down, no-change categories. The latter, concentrating only on direction of change, can be converted into estimates of means and variances, provided that one is willing to assume a specific distribution for expected price within the population sampled.

Both types of estimates have been used in the empirical analysis. As it turns out, the best results are obtained with the mean constructed by the second procedure and the variance constructed according to the first. The question of which procedure to employ is far from closed, however, and both we and others are attempting to improve upon our methods for constructing

NIA definition of personal saving is the saving concept that is analyzed.

Of the numerous equations estimated, five have been selected for presentation and discussion and are tabulated in Table 1. The first equation is included for purposes of comparison with Taylor's earlier efforts with the model based on equation (3) above. The relevant equation from Taylor (1971) is as follows: (*t*-ratios are in parentheses):

$$(4) \quad S_t = .955S_{t-1} + .449\Delta L_t + .277\Delta PR_t \\ (43.79) \quad (4.21) \quad (0.86) \\ + .893\Delta TR_t - 2.16\Delta SI_t \\ (2.86) \quad (-3.30) \\ - .901\Delta T_t + 3.65\Delta BA A_t \\ (-4.87) \quad (2.03)$$

$$R^2 = .899 \quad S_e = 2.01 \quad df = 61.$$

The period of fit for this equation was 1953-I through 1969-IV.

The most notable difference between the first equation in Table 1 and equation (4) involves the coefficients for labor income and property income. The coefficient on ΔPR is now greater than the coefficient on ΔL , which is what we expect a priori, and now has a *t*-ratio approaching 2. This emergence of property income as a somewhat energetic predictor would seem to result from recent sharp increases in farm proprietors' income. Apart from this, the coefficient on the change in payroll taxes increases, but nevertheless remains less than -1 . Note, too, that the impact of a change in transfers on saving is still substantial.

The remaining four equations in Table 1 incorporate the various uncertainty measures just discussed. Equations were estimated with ΔU as a predictor as well as U , but it was invariably weak statistically and therefore has been excluded

estimates of expected price change from the available survey information.

TABLE 1—REGRESSION RESULTS OF U.S. PERSONAL SAVING
Quarterly 1953-II—1973-IV
(*t*-ratios in parentheses)

Independent Variable	Equation				
	1	2	3	4	5
S_{t-1}	0.978 (62.60)	0.817 (13.98)	0.887 (18.87)	0.831 (14.29)	0.889 (19.06)
ΔL_t	0.245 (2.61)	0.389 (4.37)	0.430 (4.91)	0.316 (3.89)	0.409 (4.11)
ΔPR_t	0.400 (1.99)	0.266 (1.22)	0.119 (0.53)	0.469 (2.23)	0.385 (1.76)
ΔTR_t	0.774 (4.06)	0.757 (4.53)	0.861 (5.13)	0.786 (4.74)	0.865 (5.19)
ΔSI_t	-1.38 (-2.89)	-0.791 (-1.89)	-0.740 (-1.75)	-0.874 (-2.13)	-0.833 (-2.00)
ΔT_t	-0.784 (-5.43)	-0.773 (-6.23)	-0.808 (-6.53)	-0.752 (-6.15)	-0.772 (-6.25)
ΔBAA_t	3.30 (2.38)	2.67 (1.91)		3.57 (2.82)	
BAA_t		-0.738 (-1.99)		-0.959 (-2.42)	
$\Delta BAA_t Y_t$			0.0064 (2.15)		0.0082 (3.14)
$BAA_t Y_t$			-0.0022 (-3.43)		-0.0025 (-3.62)
ΔCPI_t^*		170.97 (1.74)			
$\Delta CPI_t^* Y_t$			0.495 (1.98)		
PE_t				0.492 (2.25)	
$PE_t Y_t$					0.0012 (2.22)
VPE_t		0.840 (4.57)		0.969 (5.69)	
$VPE_t Y_t$			0.0011 (3.15)		0.0015 (5.20)
U_t		0.501 (2.61)		0.510 (2.78)	
$U_t Y_t$			0.0016 (2.85)		0.0014 (2.68)
R^2	0.953	0.969	0.969	0.970	0.970
S_e	2.33	1.96	1.95	1.93	1.94
df	76	72	72	72	72

Note: The variables are defined as follows: S =personal saving; L =labor income (equals sum of wage and salary disbursements and other labor income); PR =property income (equals proprietors' income, rental income of persons, dividends, and personal interest income); TR =transfer payments; SI =personal contributions to social insurance; T =personal tax and nontax payments; BAA =yield on Baa bonds; $CPI^*=(\Delta CPI_t + \Delta CPI_{t-1})/2$; PE =mean expected price change;^a VPE =variance in mean expected price change within population sampled^a; U =civilian unemployment rate.

^a The methods of calculating PE and VPE are described in an appendix, obtainable on request from the authors.

from consideration. Equations 3 and 5 in the table have the yield on Baa bonds and the uncertainty measures interacted with the level of disposable income. The purpose of this was to allow for the possibility that the effects of uncertainty on saving vary systematically with the level of income. The results, however, do not suggest that this is the case. Finally, equations 2 and 3 differ from 4 and 5 in that 2 and 3 include the most recent six-month change in the consumer price index (*CPI*) as a predictor in place of the mean expected price change.

Addition of the uncertainty measures to the model leads to the following results and conclusions:

- (1) Except for the unemployment rate, the signs of the uncertainty measures are all as expected a priori.⁷ In particular, the results strongly support the conclusion that uncertainty regarding the future course of real and money income leads to an increase in saving.
- (2) Of the uncertainty measures considered, the strongest statistically is the variance term. As suggested by the discussion in Section II, we interpret the strong performance of this variable as representing the impact on saving of uncertainty regarding plausible upper limits to the amount of price inflation.⁸
- (3) The coefficient on payroll taxes is re-

duced considerably in absolute value and is now of the same magnitude as the coefficient on personal tax and nontax payments. Taking equation 4 as the point of reference, the only other coefficients to undergo much change (vis-à-vis equation 1) are those on S_{t-1} and labor income. The former is reduced to .83, which implies a speedier dynamic adjustment, while the latter is increased by about one-third. It is of particular interest that the inclusion of the uncertainty measures does not alter the statistical significance of the income terms. And it is especially to be noted that the coefficient on the change in transfer payments remains large.

- (4) Finally, it is to be noticed that the coefficient for the yield on Baa bonds is negative, while the coefficient for its first difference is positive. Since the factors operating on the real rate of interest should be expected to change slowly through time, the first difference in the nominal rate of interest should, in view of Irving Fisher, be dominated by price expectations. This being the case, the positive coefficient on ΔBAA is very much in keeping with the views expressed in Section II. However, the negative coefficient for *BAA*, in an expectations framework, is harder to interpret.

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⁷ The discussion in Section II viewed the unemployment rate as a proxy for (negative) transitory income, hence it was postulated that its sign should be negative. However, the fact that this is not the case suggests that the unemployment rate augments the price expectations variables as a direct measure of uncertainty.

⁸ There are, of course, other hypotheses that can be advanced to account for the importance of *VPE*. In particular, we are especially interested in exploring the possibility that *VPE* represents an aggregation effect arising from variation in the impact of price expectations on saving across income groups. Nevertheless, the fact remains that, for whatever reason, *VPE* is an important predictor empirically.

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